| EXP 10 | Performance optimization and best practices in Zoho Creator application development |
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**Aim:**

To improve the performance of the **Event Registration System** in Zoho Creator by following best practices for forms, reports, workflows and Deluge scripts.

**Problem Statement:**

As the number of events and registrations grows, the application may become slow. Reports load many records, workflows run many times, and scripts may fetch more data than needed. The application should stay fast and easy to use.

**Theory:**

Performance optimization means making the application respond quickly and use fewer resources. In Zoho Creator, the speed of an application depends on how forms, reports, workflows and Deluge scripts are designed. Best practices help to keep the app fast as data grows.

**Solution (Best Practices Used):**

- Used **Lookup** fields to link registrations with events instead of typing the event name again.
- Used **report filters, sorting and grouping** so that users see only the records they need.
- Combined the checks (participant range, event status, seat count) into **one validation workflow** instead of many workflows.
- Wrote Deluge scripts that fetch only the needed records using conditions, for example Event == input.Event && Registration_Status == "Confirmed".
- Used **Auto Number** and **Formula** fields so that values are set by the system and not by extra scripts.
- Closed an event automatically when it becomes full, so that no extra registrations are processed.
- Gave clear names to forms, fields and workflows so that the app is easy to maintain.

**Procedure:**

1. Open the Event Registration System in Zoho Creator.
2. Check the forms and make sure Lookup, Auto Number and Formula fields are used where needed.
3. Open the reports and apply filters, sorting and grouping.

4. Open the Workflow section and check that the validation logic is in one workflow.
5. Check the Deluge scripts and make sure they fetch records only with proper conditions.
6. Test the application by adding registrations and check that it responds quickly.
7. Record the observations.

**Output:**

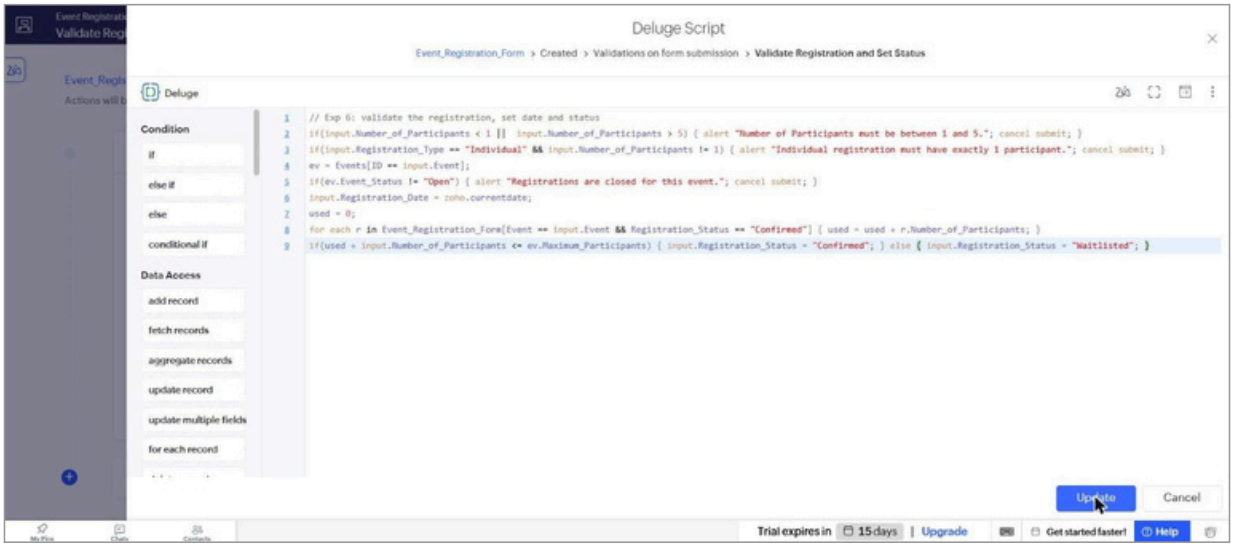



**Fig. 10.2: Single Validation Workflow with Conditional Queries**

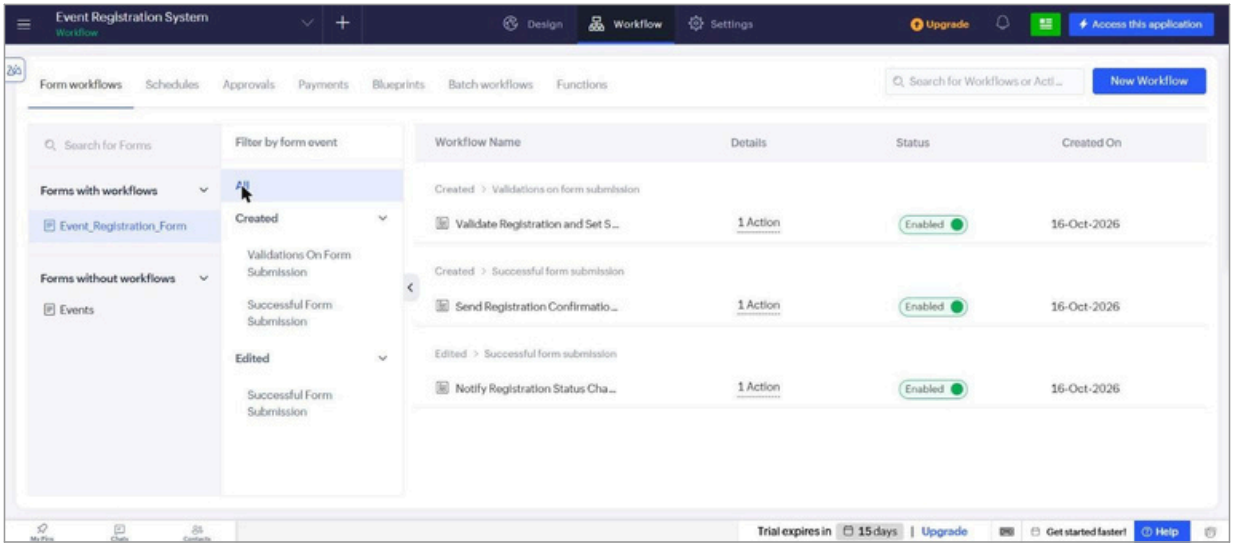



**Fig. 10.3: Small Number of Clear Workflows**

**Observation:**

| Area | Before Optimization | After Optimization |
|---|---|---|
| Reports | All records shown together | Filtered, sorted and grouped records |
| Validation | Separate checks in many places | One validation workflow |
| Data fetch | Fetching all registrations | Fetching only confirmed records of one event |
| Event status | Closed by hand | Closed automatically when full |
| Maintenance | Hard to find logic | Clear names and fewer workflows |

**Result:**

The performance of the Event Registration System was improved by using filters, combined workflows, conditional queries and system fields. The application responds quickly and is easy to maintain.

**Conclusion:**

Following best practices in Zoho Creator keeps an application fast, simple and ready to handle more data in the future.